



GATE CS / IT 2027

Database Management Systems

Quick Revision Sheet

Complete GATE syllabus coverage — All topics and subtopics

Normalization

Normal Forms

- 1NF: atomic values, no repeating groups
- 2NF: 1NF + no partial dependency on primary key
- 3NF: 2NF + no transitive dependency (non-key $\rightarrow$ non-key)
- BCNF: every determinant is a superkey (stricter than 3NF)
- 4NF: BCNF + no non-trivial multi-valued dependencies

Armstrong's Axioms & Closure

- Reflexivity: $Y \subseteq X \Rightarrow X \rightarrow Y$
- Augmentation: $X \rightarrow Y \Rightarrow XZ \rightarrow YZ$
- Transitivity: $X \rightarrow Y$ and $Y \rightarrow Z \Rightarrow X \rightarrow Z$
- Closure X^+ : all attributes functionally determined by X
- Minimal cover: smallest equivalent set of FDs

SQL & Relational Algebra

Relational Algebra

- Select (σ): filter rows by condition
- Project (π): select specific columns
- Join ($\Join$): combine tuples from two relations
- Natural Join: join on all common attributes and project
- Division: find tuples related to ALL tuples in other relation
- Rename (ρ): rename relation or attributes

SQL Tips

- WHERE filters rows, HAVING filters groups
- GROUP BY comes before HAVING
- NULL in comparisons yields UNKNOWN
- EXISTS returns true if subquery has any rows
- DISTINCT removes duplicate rows in result
- Subquery in FROM is called derived table

Transactions & Concurrency

ACID & Serializability

- Atomicity: all-or-nothing execution
- Consistency: DB constraints always satisfied
- Isolation: concurrent transactions appear serial
- Durability: committed data survives system failure
- Conflict Serializability: precedence graph must be acyclic
- Two-Phase Locking (2PL): growing then shrinking phase

Indexing

Index Type	Description	Best For
Dense	One entry per record	Exact match queries
Sparse	One entry per block	Sequential/clustered data
Primary	On primary key, clustered	Sorted key lookups
Secondary	On non-key attribute	Multi-attribute queries
B+ Tree	Balanced tree, data in leaves	Range queries, most common
Hash Index	Hash function on key	Exact match only